

Introduction to Stochastic Actor-Oriented Models

Fundamentals of SAOMs

Johan Koskinen

Department of Statistics
Stockholm University
University of Melbourne

March 19, 2026



Stockholm
University

Preamble

- All material is on the workshop repository
<https://github.com/johankoskinen/CHDH-SNA>
 - ▶ Download the RMarkdown file CHDH-SNA-3.Rmd
- In order to run the Markdown you need
 - ▶ The R-package 
 - ▶ The RStudio interface  Studio
- We will predominantly use the packages
 - ▶ sna
 - ▶ network
 - ▶ RSiena



Outline of workshops

- ① (Basic) Introduction to SAOM
 - ▶ SAOM as an agent-based model
 - ▶ How to estimate a SAOM
- ② (Social Influence) Analysing social influence with SAOM
 - ▶ Accounting for nodal attributes
 - ▶ Modelling change of nodal attributes
 - ▶ Trouble shooting and dealing with common issues
- ③ Advanced topics in SAOM
 - ▶ Even more types of data
 - ▶ Likelihood-based estimation
 - ▶ Settings and imperfect data
 - ▶ Modelling multiple parallel networks



What type of data do we want to explain: adjacency matrix

Data represented as adjacency matrices

$$\mathbf{x} = \begin{pmatrix} . & 0 & 0 & 0 & 1 \\ 1 & . & 0 & 0 & 0 \\ 1 & 1 & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 \\ 0 & 0 & 1 & 1 & . \end{pmatrix}$$

where $x_{ij} = 1$ or 0 according to whether $i \rightarrow j$ or not.

What type of data do we want to explain: longitudinal

Data represented as adjacency matrices
where elements **change**

$$x(t_0) = \begin{pmatrix} . & 0 & 0 & 0 & 1 \\ 1 & . & 0 & 0 & 0 \\ 1 & 1 & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 \\ 0 & 0 & 1 & 1 & . \end{pmatrix}$$

What type of data do we want to explain

Data represented as adjacency matrices
where elements **change**

$$x(t_1) = \begin{pmatrix} . & 1 & 0 & 0 & 1 \\ 1 & . & 0 & 0 & 0 \\ 1 & 0 & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 \\ 1 & 0 & 1 & 1 & . \end{pmatrix}$$

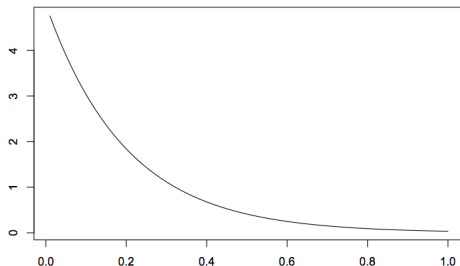
What type of data do we want to explain

Data represented as adjacency matrices
where elements **change**

$$x(t_2) = \begin{pmatrix} . & 1 & 0 & \mathbf{1} & 1 \\ 1 & . & 0 & 0 & \mathbf{1} \\ 1 & \mathbf{1} & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 \\ 1 & 0 & \mathbf{0} & 1 & . \end{pmatrix}$$

SAOM: the rate of change

At random points in time, at rates λ_i



nodes/individuals/**actors** are given **opportunities to change**

SAOM: the *direction* of change

Conditional on an actor having an **opportunity for change**
the probability for each outcome

- ⊙ is modelled like multinomial logistic regression
- ⊙ reflects the attractiveness of the outcome to the actor

Micro-step

When actor i has opportunity to change

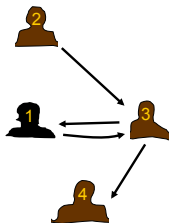
They may toggle x_{ij} to $1 - x_{ij}$

We call the new network

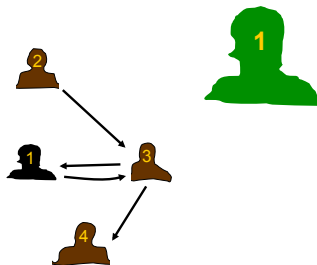
$$X(i \rightsquigarrow j)$$

the network X that *differs* in exactly **one** tie-variable x_{ij}

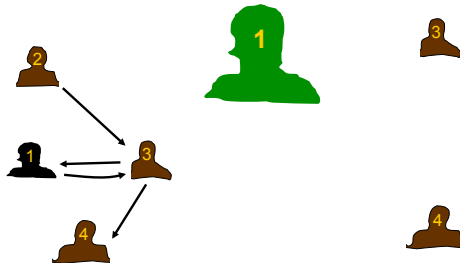
$$x = \begin{array}{|c|c|c|c|} \hline - & 0 & 1 & 0 \\ \hline 0 & - & 1 & 0 \\ \hline 1 & 0 & - & 1 \\ \hline 0 & 0 & 0 & - \\ \hline \end{array}$$



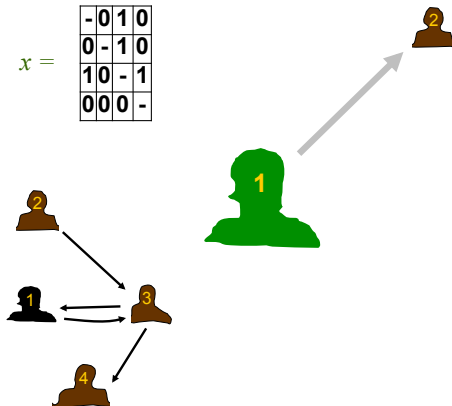
$$x = \begin{array}{|c|c|c|c|} \hline - & 0 & 1 & 0 \\ \hline 0 & - & 1 & 0 \\ \hline 1 & 0 & - & 1 \\ \hline 0 & 0 & 0 & - \\ \hline \end{array}$$



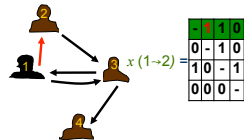
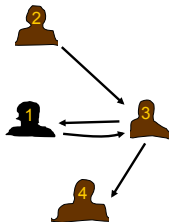
$$x = \begin{bmatrix} -0 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$



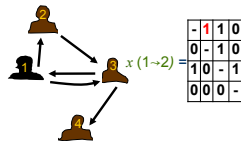
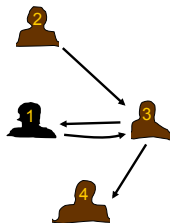
$$x = \begin{bmatrix} -0 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$



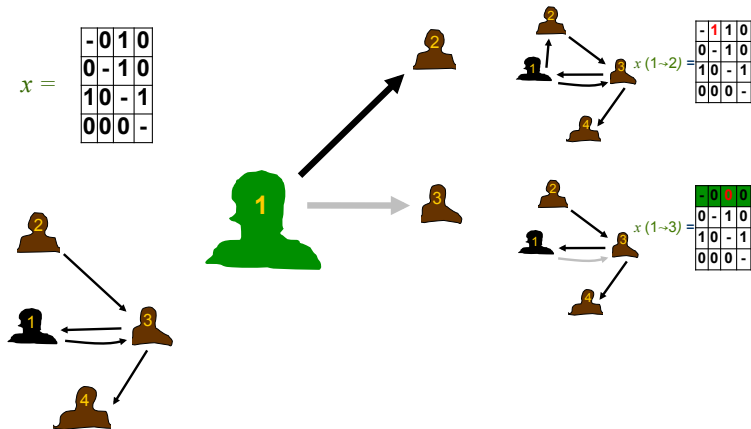
$$x = \begin{bmatrix} - & 0 & 1 & 0 \\ 0 & - & 1 & 0 \\ 1 & 0 & - & 1 \\ 0 & 0 & 0 & - \end{bmatrix}$$

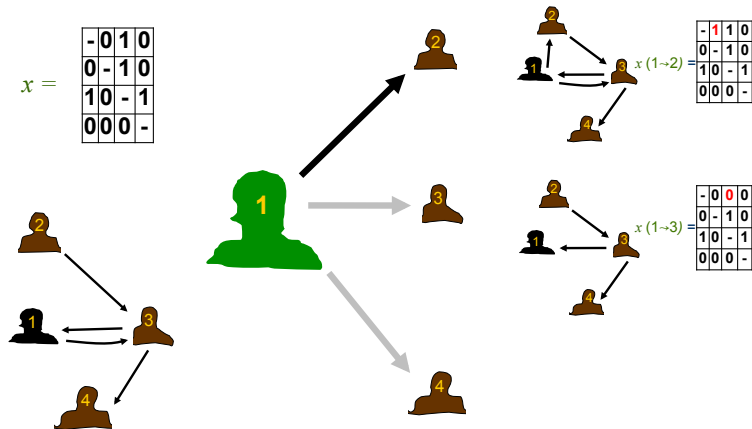


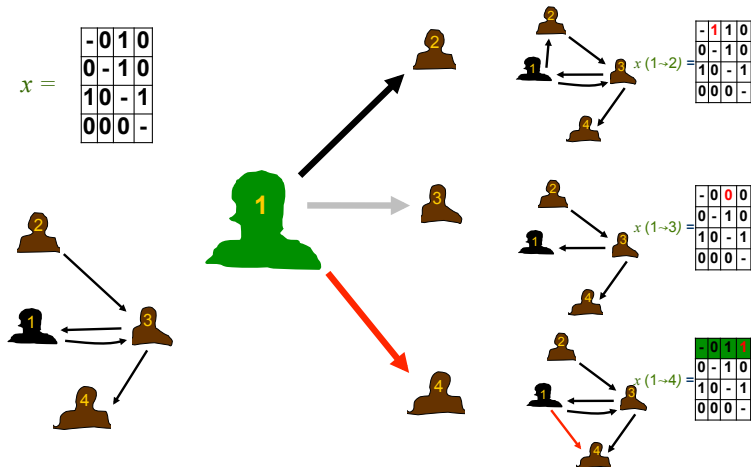
$$x = \begin{bmatrix} -0 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

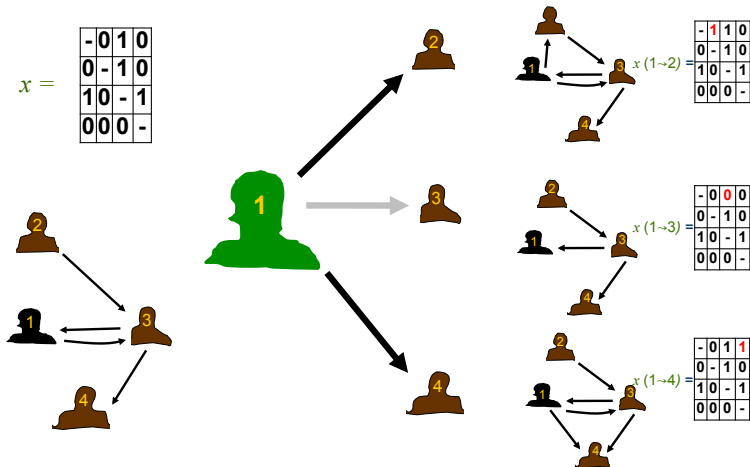


$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$









Probability of a change

Of the three changes (for $j = 2, 3, 4$) available to i (here 1) the probability that i toggles the tie $i \rightarrow j$ is given by

One-step jump probability

$$p_{ij}(\beta, \underbrace{X}_{\text{current network}}) = \frac{\exp(f_i(\beta, \underbrace{X(i \rightsquigarrow j)}_{\text{new network}}))}{\underbrace{\sum_{h=1}^n \exp(f_i(\beta, X(i \rightsquigarrow h)))}_{\text{all possible changes}}},$$

where

- $X(i \rightsquigarrow j)$ is the network resulting from the change
- β are **statistical parameters**
- f_i describes the attractiveness of $X(i \rightsquigarrow j)$ to i

Probability of a change: utility

One-step jump probability: *can* be derived as:

- Each network X has **utility** $U_i(X, t)$ for i
- Actor i chooses network X that maximises $U_i(X, t)$

If (random) utility has form

$$U_i(X, t) = \underbrace{f_i(\beta, X)}_{\text{objective function}} + \epsilon_{it}.$$

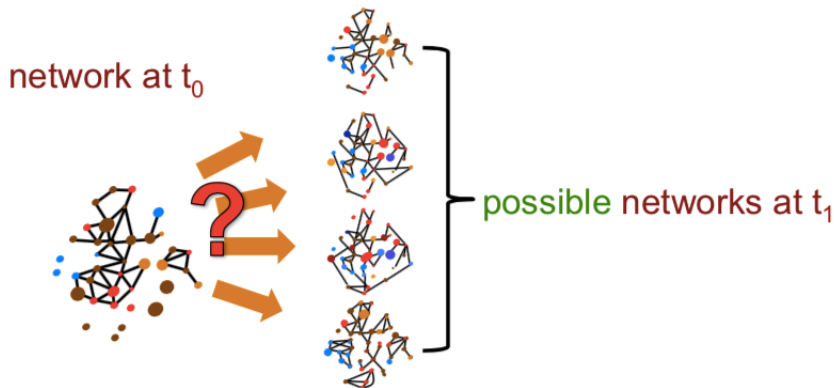
↑↑

random component

Actors are (*myopically*) maximising the utility of their network ties



Agent-based: Change driven by incremental updates

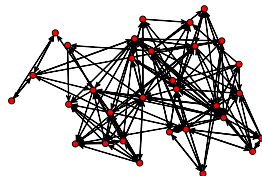
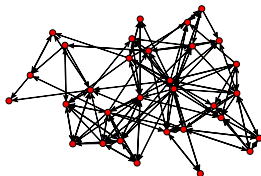


van de Bunt data set: R

Markdown: <https://raw.githubusercontent.com/johankoskinen/CHDH-SNA/main/Markdowns/CHDH-SNA-4.Rmd>

```
library('RSiena')  
library('network')  
library('sna')  
tmp3[is.na(tmp3)] <- 0 # remove missing  
tmp4[is.na(tmp4)] <- 0 # remove missing  
par(mfrow = c(1,2))  
coordin <- plot(as.network(tmp3))  
plot(as.network(tmp4),coord=coordin)
```





Let us assume that i **ONLY** cares about not having too many or too few ties:

$$f_i(\beta, X) = \beta \sum_j x_{ij}$$

meaning that

$$p_{ij}(\beta, X) = \frac{\exp\{\beta(1 - 2x_{ij})\}}{\sum_{h=1}^n \exp\{\beta(1 - 2x_{ih})\}},$$

because if

- currently $x_{ij} = 1$, then
- the number of ties for i in $X(i \rightsquigarrow j)$ will be one less (-1) ,
- and if currently $x_{ij} = 0 \dots$

Simulation settings: actors only care about degree

Let the rate be equal for all $\lambda_i = \lambda = 5.7288$

- ✓ in each iteration, actor with shortest waiting time 'wins' (and gets to change)
- ✓ on average every actor gets 5.7 opportunities to change

and set $\beta = -0.7349$

- ✓ if $\beta = 0$ actor would not care if tie was added or deleted
- ✓ here $\beta < 0$ meaning that actor wants less than half of the possible ties



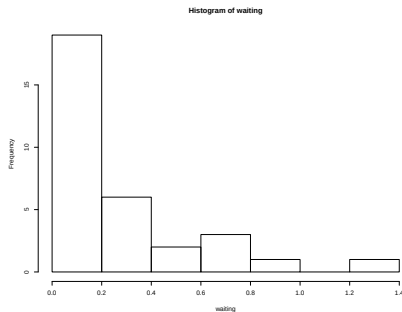
van de Bunt data set

```
myNet1 <- sienaDependent(array(c(tmp3, tmp4),  
                               dim=c(32, 32,2)))  
mydata <- sienaDataCreate(myNet1)  
myeff <- getEffects(mydata)  
myeff <- includeEffects(myeff, recip, include=FALSE)  
myeff$initialValue[  
    myeff$shortName == 'Rate'] <- 5.7288  
myeff$initialValue[  
    myeff$shortName=='density'] [1] <- -0.7349
```

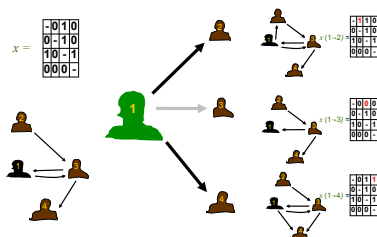


Model: rate

```
waiting <- rexp(32,5.7288)
hist(waiting)
which( waiting == min(waiting)) ← the winner
```



Model: conditional one-step change probability



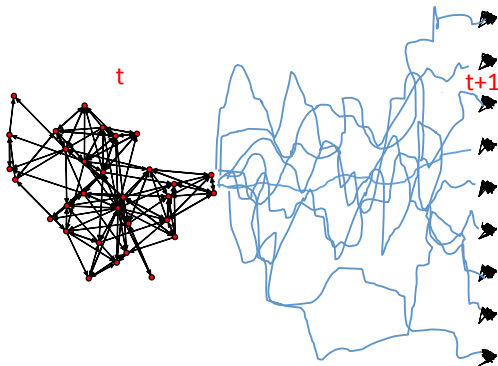
$$\Pr(1 \leadsto 2) = \frac{e^{-0.7349}}{e^{-0.7349} + e^{1.1059} + e^{-0.7349} + 1} = 0.1185$$

$$\Pr(1 \leadsto 3) = \frac{e^{1.1059}}{e^{-0.7349} + e^{1.1059} + e^{-0.7349} + 1} = 0.5156$$

$$\Pr(1 \leadsto 4) = \frac{e^{-0.7349}}{e^{-0.7349} + e^{1.1059} + e^{-0.7349} + 1} = 0.1185$$

Of course, in van de Bunt every actor has 31+1 choices for change

Now let us simulate



Simulate from t_0 to t_1 (`simOnly = TRUE`)

```
sim_model <- sienaAlgorithmCreate(
  projname = 'sim_model',
  cond = FALSE,
  useStdInits = FALSE, nsub = 0 ,
  simOnly = TRUE)
sim_ans <- siena07( sim_model, data = mydata,
  effects = myeff,
  returnDeps = TRUE, batch=TRUE )
```

The object `sim_ans` will now contain 1000 simulated networks



Extract networks

```
n <- dim(tmp4)[1]  
mySimNets <- reshapeRSienaDeps( sim_ans , n )
```

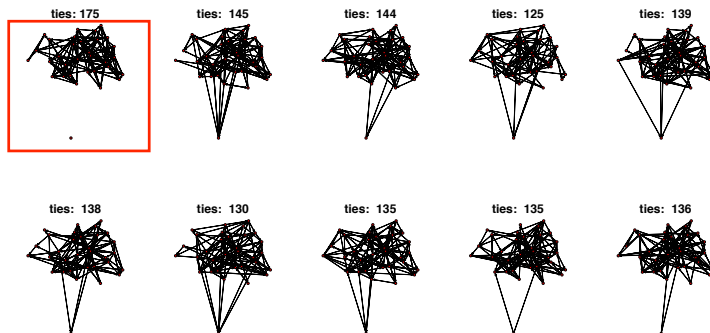
The object `mySimNets` is a 1000 by n by n array of adjacency matrices

Plot observed network and 9 simulated

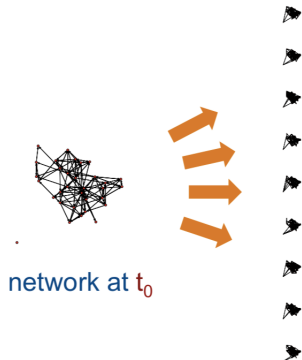
```
pdf(file='simnets1.pdf', width = 9,height =4.5)
par(mfrow=c(2,5), oma = c(0,4,0,0) + 0.1,
    mar = c(5,0,1,1) + 0.1)
plot(as.network(tmp4),coord=coordin,
      main=paste('ties:',sum(tmp4) ) )
apply(mySimNets[1:9,,],1,function(x)
      plot(as.network(x),
            coord=coordin,
            main=paste('ties: ',sum(x))) )
dev.off()
```



The **observed** at t_1 and possible networks at t_1



Simulated networks v t_1 obs



Dyad Census

```
> dyad.census(tmp4)
      Mut Asym Null
[1,]  46   83  367
> dyad.census(mySimNets[1:9,,])
      Mut Asym Null
[1,]   23  128  345
[2,]   25  133  338
[3,]   17  136  343
[4,]   20  134  342
[5,]   16  143  337
[6,]   21  136  339
[7,]   26  118  352
[8,]   23  128  345
[9,]   30  122  344
```



Conclusions

A process where i ONLY cares about not having too many or too few ties does to replicate the reciprocity at t_1

Assume that i ALSO cares about having ties $i \rightarrow j$ reciprocated $j \rightarrow i$

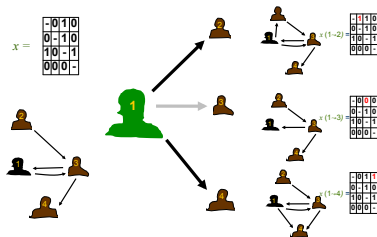
$$f_i(\beta, X) = \beta_d \sum_j x_{ij} + \beta_r \sum_j x_{ij} x_{ji}$$

meaning that probability that i toggles relationship to j

$$p_{ij}(\beta, X) = \frac{\exp \{ \beta_d (1 - 2x_{ij}) + \beta_r (1 - 2x_{ij}) x_{ji} \}}{\sum_{h=1}^n \exp \{ \beta_d (1 - 2x_{ih}) + \beta_r (1 - 2x_{ih}) x_{hi} \}},$$



Model



Objective function:

$$\beta_d \sum_j x_{ij} + \beta_r \sum_j x_{ij} x_{ji}$$

- adding $1 \rightarrow 2$: β_d
- deleting $1 \rightarrow 3$: $-\beta_d - \beta_r$
- adding $1 \rightarrow 4$: β_d

Our simulated networks had too **few** reciprocated dyads so we need to set $\beta_r \dots$

Simulation settings: actors care about degree and reciprocity

Let the rate be equal for all $\lambda_i = \lambda = 6.3477$

- ✓ on average every actor gets 6.3 opportunities to change

and set $\beta_d = -1.1046$

- ✓ here $\beta_d < 0$ - actors do not want too many ties

and set $\beta_r = 1.2608$

- ✓ here $\beta_r > 0$ - actors prefer reciprocated to assymmetric ties

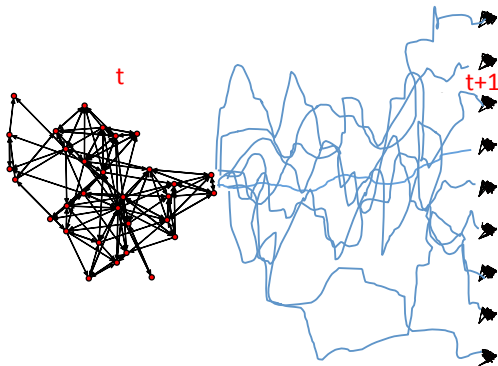


van de Bunt data set

```
myeff <- includeEffects(myeff, recip, include=TRUE)
myeff$initialValue[
    myeff$shortName == 'Rate'] <- 6.3477
myeff$initialValue[
    myeff$shortName == 'density'][1] <- -1.1046
myeff$initialValue[
    myeff$shortName == 'recip'][1] <- 1.2608
```



Now let us simulate



Simulate from t_0 to t_1 now with reciprocity (`simOnly = TRUE`)

```
sim_model <- sienaAlgorithmCreate(
  projname = 'sim_model',
  cond = FALSE,
  useStdInits = FALSE, nsub = 0 ,
  simOnly = TRUE)
sim_ans <- siena07( sim_model, data = mydata,
  effects = myeff,
  returnDeps = TRUE, batch=TRUE )
```

The object `sim_ans` will now contain 1000 simulated networks
NOTE: this piece of code is unchanged

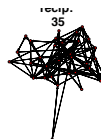
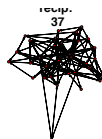
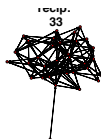
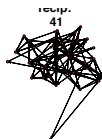
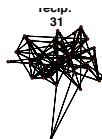
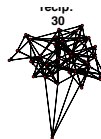
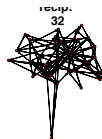
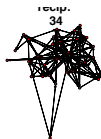
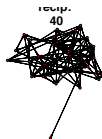


Plot observed network and 9 simulated

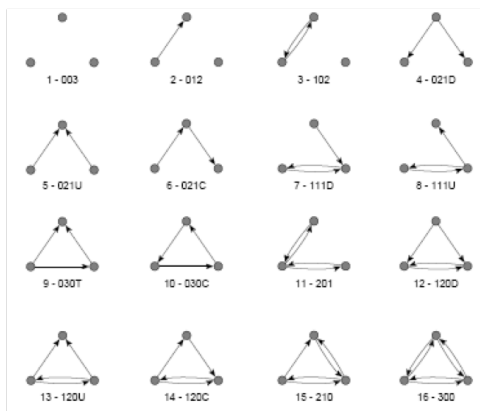
```
mySimNets <- reshapeRSienaDeps(sim_ans,n)
plot(as.network(tmp4),coord=coordin,
      main=paste('recip:',dyad.census(tmp4)[1] ) )
apply(mySimNets[1:9,,],1,function(x)
      plot(as.network(x),
            coord=coordin,
            main=paste('recip:
',dyad.census(x)[1] ) ) )
```



The **observed** at t_1 and possible networks at t_1



Simulated networks v t_1 obs: triad census



Simulated networks v t_1 obs: triad census

```
> triad.census(tmp4)
      003  012 102 021D 021U 021C 111D 111U 030T 030C 201 120D 120U 120C 210 300
[1,] 2078 1329 745 146  80  52  65 217  37  0 68 16 65 10 30 22
> triad.census(mySimNets[1:9,,])
      003  012 102 021D 021U 021C 111D 111U 030T 030C 201 120D 120U 120C 210 300
[1,] 1968 1381 718  95  84 160 148 224 16  4 77 13 13 17 33  9
[2,] 2067 1348 703  85  82 154 150 191 16  8 75  9 11 25 27  9
[3,] 2073 1397 687 102  75 158 129 181 18  2 68 14 13 23 13  7
[4,] 2185 1313 733  78  60 132 102 172 20  7 89  7 10 20 28  4
[5,] 2040 1340 766  89  64 155 129 189 18  7 82 12 11 19 27 12
[6,] 2206 1403 669  76  68 135 122 143 17  6 64  8 12  9 17  5
[7,] 1788 1357 760 113  89 169 195 238 30 11 98 14 16 34 37 11
[8,] 2164 1301 681  70  65 136 168 174 12  5 91 15  8 30 28 12
[9,] 1988 1383 729 111  67 151 148 173 26 10 82 13 22 19 32  6
```

Reciprocity is clearly not enough to explain the incidence of *transitive triangles* and *simmelian ties* (3 Mutual 0, Assymetric, 0 Null)

Assume that i ALSO cares about *closure*

$$f_i(\beta, X) = \exp \left\{ \beta_d \sum_j x_{ij} + \beta_r \sum_j x_{ij} x_{ji} + \beta_t s_{i,t}(x) \right\}$$

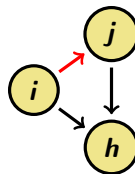
Modelled through, e.g.

transitive triplets effect,

number of transitive patterns in i 's ties

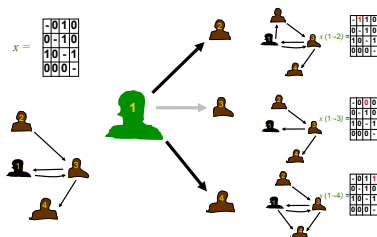
$(i \rightarrow j, j \rightarrow h, i \rightarrow h)$

$$s_{i,t}(x) = \sum_{j,h} x_{ij} x_{jh} x_{ih}$$



transitive triplet

Model



Objective function including $s_{i,t}(x)$:

- adding 1 → 2: $\dots + \beta_t$
- deleting 1 → 3: no change in closure
- adding 1 → 4: $\dots + \beta_t$

Our simulated networks had too **few** 030T and 300 so we need to set $\beta_t \dots$

Assume actors care about degree, reciprocity, and closure

Let the rate be equal for all $\lambda_i = \lambda = 7.0959$

✓ on average every actor gets 7 opportunities to change

and set $\beta_d = -1.6468$

✓ here $\beta_d < 0$ - actors do not want too many ties

and set $\beta_r = 0.8932$

✓ here $\beta_r > 0$ - actors prefer reciprocated to asymmetric ties

and set $\beta_t = 0.2772$

✓ here $\beta_t > 0$ - actors prefer ties that close open triads

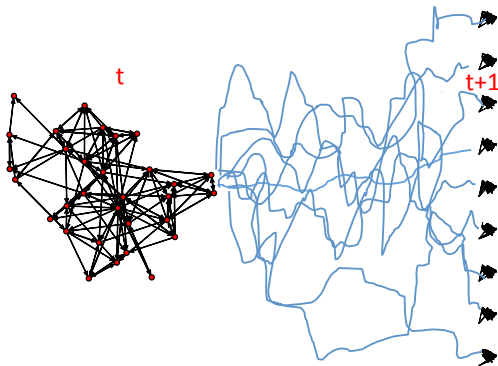


van de Bunt data set

```
myeff <- includeEffects(myeff, recip, include=TRUE)
myeff <- includeEffects(myeff, transTrip, include=TRUE)
myeff$initialValue[
    myeff$shortName == 'Rate'] <- 7.0959
myeff$initialValue[
    myeff$shortName == 'density'][1] <- 1.6468
myeff$initialValue[
    myeff$shortName == 'recip'][1] <- 0.8932
myeff$initialValue[
    myeff$shortName == 'transTrip'][1] <- 0.2772
```



Now let us simulate



Simulate from t_0 to t_1 now with transitivity (`simOnly = TRUE`)

```
sim_model <- sienaAlgorithmCreate(
  projname = 'sim_model',
  cond = FALSE,
  useStdInits = FALSE, nsub = 0 ,
  simOnly = TRUE)
sim_ans <- siena07( sim_model, data = mydata,
  effects = myeff,
  returnDeps = TRUE, batch=TRUE )
```

NOTE: this piece of code is unchanged

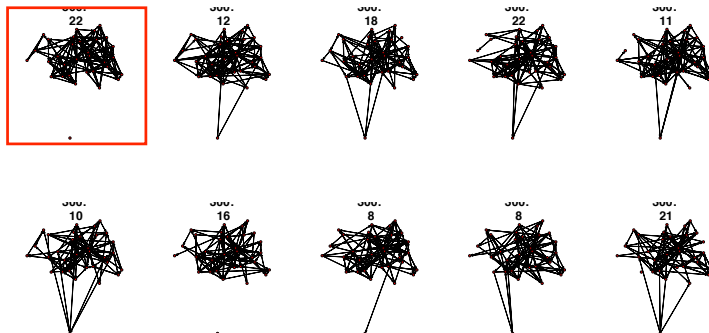


Plot observed network and 9 simulated

```
mySimNets <- reshapeRSienaDeps(sim_ans,n)
plot(as.network(tmp4),coord=coordin,
      main=paste('recip:',triad.census(tmp4)[16] ) )
apply(mySimNets[1:9,,],1,function(x)
      plot(as.network(x),
            coord=coordin,
            main=paste('300:
',triad.census(x)[16] ) ) )
```



The **observed** at t_1 and possible networks at t_1



Simulated networks v t_1 obs: triad census

```
> triad.census(tmp4)
      003  012 102 021D 021U 021C 111D 111U 030T 030C 201 120D 120U 120C 210 300
[1,] 2078 1329 745 146   80   52   65  217   37   0  68   16   65   10   30  22
> triad.census(mySimNets[1:9,,])
      003  012 102 021D 021U 021C 111D 111U 030T 030C 201 120D 120U 120C 210 300
[1,] 2029 1223 662  108   79  159  138  257   36   6  82   19   35   29  68  30
[2,] 2163 1292 727   89   62  125  136  160   15   4  61   9   23   25  29  40
[3,] 2551 1191 657   44   34   84   88  143    5   0  78   6   16   14  27  22
[4,] 1990 1430 576  100   84  173  128  200   23  10 103   12   38   22  45  26
[5,] 2219 1390 631  103   83  121   84  175   25   5  45   8   21   9  31  10
[6,] 2031 1218 799   75   56  109  136  213   15   9 114   15   25   25  61  59
[7,] 2079 1443 597  104   72  160   97  206   24   4  54   17   24   24  38  17
[8,] 2105 1256 764   77   55  114   99  212   12   5  98   11   33   16  50  53
[9,] 2405 1260 569  103   55  126   78  184   22   6  41   12   25   8  46  20
```

Reciprocity together with transitivity seems enough to explain the incidence of *transitive triangles* and *simmelian ties* (3 Mutual 0, Assymmetric, 0 Null)

for arbitrary rate function $\lambda_i(\alpha, \rho, x)$

- 1 Set $t = 0$ and $x = X(0)$.
- 2 Generate S according to the exponential distribution with mean $1/\lambda_+(\alpha, \rho, x)$ where $\lambda_+(\alpha, \rho, x) = \sum_i \lambda_i(\alpha, \rho, x)$.
- 3 Select $i \in \{1, \dots, n\}$ using probabilities $\frac{\lambda_i(\alpha, \rho, x)}{\lambda_+(\alpha, \rho, x)}$.

- ④ Select $j \in \{1, \dots, n\}$, $j \neq i$ using probabilities $p_{ij}(\beta, x)$.
- ⑤ Set $t = t + S$ and $x = x(i \rightsquigarrow j)$.
- ⑥ Go to step 2
(unless stopping criterion is satisfied).

Questions?

What effects are there?

- RSiena Manual

http://www.stats.ox.ac.uk/~snijders/siena/RSiena_Manual.pdf -
check for shortName

- scroll through the effects available to you for your data **myeff** - check for shortName
- also **effectsDocumentation(myeff)**

Where did I get these numbers?



Estimation



Estimation by Method of Moments: data

Basics for data

- You need at least 2 observations on $X(t)$ for waves t_0, t_1
- First observations is fixed and contains no information about θ
- No assumption of a stationary network distribution



Estimation by Method of Moments: procedure

How to estimate $\theta = (\lambda, \beta)$?

- pick starting values for θ
- simulate from $X(t_0)$ until t_1 - call the simulated network (-s) X_{rep}
- if statistic $Z_k(X_{\text{rep}})$ for parameter k is different to $Z_k(X_{\text{obs}})$, adjust accordingly



Estimation by Method of Moments: aim

For suitable statistic $Z = (Z_1, \dots, Z_K)$,
i.e., K variables which can be calculated from the network;
the statistic Z_k must be *sensitive* to the parameter θ_k
e.g. number of mutual dyads is sensitive to the reciprocity parameter (as we have seen)

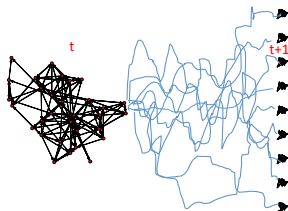
The MoM estimate is a value: $\hat{\theta}$ of θ such that for

- observed stats $Z(X_{\text{obs}})$
- and the expected value $E_{\theta}(Z(X_{\text{rep}}))$

$$E_{\hat{\theta}} \{Z(X_{\text{rep}})\} = Z(X_{\text{obs}}) .$$



Method of Moments matches the moments



Do we have to do this for every update of the parameter θ ?

Robbins-Monro algorithm

The moment equation $E_{\hat{\theta}}\{Z\} = z$ cannot be solved by analytical or the usual numerical procedures

Stochastic approximation (Robbins-Monro, 1951)

Iteration step:

$$\hat{\theta}_{N+1} = \hat{\theta}_N - a_N D^{-1}(z_N - z) , \quad (1)$$

where z_N is a simulation of Z with parameter $\hat{\theta}_N$,

D is a suitable matrix, and $a_N \rightarrow 0$.

Computer algorithm has 3 phases:

- ① brief phase for preliminary estimation of $\partial E_{\theta} \{Z\} / \partial \theta$ for defining D ;
- ② estimation phase with Robbins-Monro updates, where a_N remains constant in *subphases* and decreases between subphases;
- ③ final phase where θ remains constant at estimated value; this phase is for checking that

$$E_{\hat{\theta}} \{Z\} \approx z ,$$

and for estimating D_{θ} and Σ_{θ} to calculate standard errors.



Convergence

We say that $E_{\hat{\theta}}\{Z\} = z$ is approximately satisfied if, for each statistic $Z_k(X_{\text{obs}})$ is within 0.1 standard deviation of $E_{\theta}(Z(X_{\text{rep}}))$.
This is provided in the output as the *convergence t-ratio* (and the overall maximum convergence ratio is less than 0.25)



Summary

What is the purpose of having the embedded Markov Chain in continuous time?

DYNAMICS

can model change of *tie* as dependent on current ties AND behaviour

can model change in *behaviour* as dependent on current behaviour AND the behavior of those you are tied to



Summary

What is the purpose of having the embedded Markov Chain in continuous time?

STATISTICAL

This is a statistical model that has **estimable parameters** for selection and influence

This is a **generative model** from which we can also generate replicate data AND assess GOF



Halfway Summary

What is the purpose of having the embedded Markov Chain in continuous time?

STATISTICAL

This is a statistical model that has **estimable parameters** for selection and influence

This is a **generative model** from which we can also generate replicate data AND assess GOF

Compare

- Generalized Estimation Equations
- Regressing behaviour wave 1 on wave 0



Goodness-of-fit

Testing assumptions: Goodness-of-fit (GOF)

We can (almost) always get estimates
but model is very complex
so how do we know that it is realistic?



Two routines for goodness-of-fit

- `sienaTimeTest()`
for testing time heterogeneity
- `sienaGOF()`
for checking that the model reproduces the features of the observed networks (that were not modelled).



Time-test

Standard assumptions M waves, the $M - 1$ periods follow the same model with the **same** parameters.

Use

- `sienaTimeTest()`
to test if some parameters differ across any of the periods
- if test 'positive'
include interactions with time using
`includeTimeDummy()`

see `RscriptSienaTimeTest.r`



Extension 2: Is model homogenous over time

Example time test (Lospinoso et al., 2010)

vdb_tt.R:

```
vdb.ans1 <- siena07(vdb.model, data=vdb.data,  
                   effects=vdb.eff,  
                   useCluster=FALSE, initC=TRUE)  
timetest.1 <- sienaTimeTest(vdb.ans1)  
summary(timetest.1)  
plot(timetest.1, effects=c(1,3))
```



Goodness of fit

Principle: simulate replicate data
and check how simulations compare to observed data
This is exactly what we did in 'Simulating SAOM'
What are we looking for?
does model capture features that we have not modelled?



built in GOF-function

Siena has function `sienaGOF()`

This operates on your `siena`-object

generated from `siena07()` with option `returnDeps = TRUE`



choosing features for GOF

Some preprogrammed 'auxiliary' functions
that can be passed to `sienaGOF` are:

`OutdegreeDistribution()`

`IndegreeDistribution()`

`BehaviorDistribution()`

you can also create custom functions



More help on GOF

Use ? function and sienaGOF_new.R

```
results1 <- siena07(myalg, data=mydata,  
                  effects=myeff, returnDeps=TRUE)  
gof1.od <- sienaGOF(results1, verbose=TRUE,  
                  varName="friendship",  
                  OutdegreeDistribution,  
                  cumulative=TRUE, levls=0:10)  
  
gof1.od  
plot(gof1.od)
```

See example script

https://www.stats.ox.ac.uk/~snijders/siena/sienaGOF_vdB.R



Modelling behaviour change - social influence



Change to BEHAVIOUR



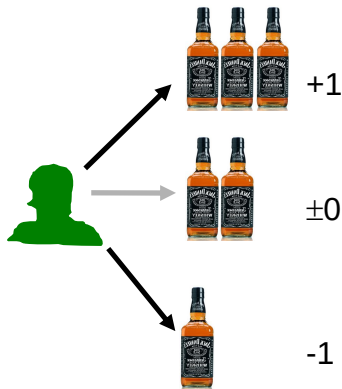
Satisfaction with new state: $f_i +$ random component

Change to BEHAVIOUR



Satisfaction with new state: $f_i +$ random component

Change to BEHAVIOUR



Satisfaction with new state: $f_i +$ random component

For the behaviours, the formula of the change probabilities is

$$p_{ihv}(\beta, z) = \frac{\exp(f(i, h, v))}{\sum_{k,u} \exp(f(i, k, u))}$$

where $f(i, h, v)$ is the objective function calculated for the potential new situation after a behaviour change,

$$f(i, h, v) = f_i^z(\beta, z(i, h \rightsquigarrow v)) .$$

Again, multinomial logit form.

Things that go into the objective functions - selection

Homophily effects:

counts of the number of ties to people that are “like me”



Things that go into the objective functions - influence

Controls:

- 1 Gender
- 2 Age
- 3 Education

For influence
effects:
imitation
persuasion
etc



Things that go into the objective functions - influence

Controls:

- 1 Gender
- 2 Age
- 3 Education

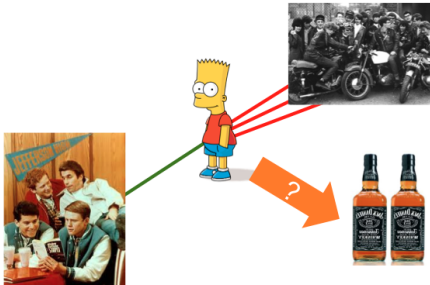


For **influence**
effects:
imitation
persuasion
etc

Things that go into the objective functions - influence

Controls:

- 1 Gender
- 2 Age
- 3 Education



For **influence** effects:
imitation
persuasion
etc

Example: 50 girls in a Scottish secondary school

Study of smoking initiation and friendship (starting age 12-13 years)
(following up on earlier work by P. West, M. Pearson & others).
with sociometric & behavior questionnaires at three moments, at appr. 1
year intervals.

Smoking: values 1-3;

drinking: values 1-5;

covariates:

gender, smoking of parents and siblings (binary),

money available (range 0-40 pounds/week).



Rename data that was automatically loaded

```
friend.data.w1 <- s501
friend.data.w2 <- s502
friend.data.w3 <- s503
drink <- s50a
smoke <- s50s
friendshipData <- array( c( friend.data.w1,
                             friend.data.w2,
                             friend.data.w3 ),
                          dim = c( 50, 50, 3 ) )
```



Define dependent/independent data

```
friendship <- sienaDependent(friendshipData)
drinking <- sienaDependent( drink, type = "behavior" )
smoke1 <- coCovar( smoke[ , 1 ] )
```



Join data and get effects

```
NBdata <- sienaDataCreate( friendship,  
                           smoke1,  
                           drinking )  
NBeff <- getEffects( NBdata )
```



Define structural network effects

```
NBeff <- includeEffects( NBeff, transTrip, transRecTrip )
```

Define covariate effects on the network (selection)

```
NBeff <- includeEffects( NBeff,  
                        egoX, egoSqX, altX, altSqX,  
                        diffSqX,  
                        interaction1 = "drinking" )  
NBeff <- includeEffects( NBeff, egoX, altX, simX,  
                        interaction1 = "smoke1" )
```

Define effects on drinking (influence)

```
NBeff <- includeEffects( NBeff, avAlt, name="drinking",  
                        interaction1 = "friendship" )
```

Define estimation settings and estimate

```
myalgorithm1 <- sienaAlgorithmCreate( projname = 's50_NB' )  
NBans <- siena07( myalgorithm1,  
                  data = NBdata, effects = NBeff,  
                  returnDeps = TRUE )
```



Result selection

Effect	par.	(s.e.)	t stat.
constant friendship rate (period 1)	6.21	(1.08)	-0.0037
constant friendship rate (period 2)	5.01	(0.87)	0.0042
outdegree (density)	-2.82	(0.27)	-0.0809
reciprocity	2.82	(0.35)	0.0559
transitive triplets	0.90	(0.16)	0.0741
transitive recipr. triplets	-0.52	(0.24)	0.0695
smoke1 alter	0.07	(0.17)	0.0343
smoke1 ego	-0.00	(0.15)	0.0747
smoke1 similarity	0.25	(0.24)	0.0158
drinking alter	-0.06	(0.15)	0.0158
drinking squared alter	-0.11	(0.14)	0.0704
drinking ego	0.04	(0.13)	0.0496
drinking squared ego	0.22	(0.12)	0.0874
drinking diff. squared	-0.10	(0.05)	0.0583

convergence t ratios all < 0.09 .

Overall maximum convergence ratio 0.19.



Result Influence

Effect	par.	(s.e.)	<i>t</i> stat.
rate drinking (period 1)	1.31	(0.34)	-0.0692
rate drinking (period 2)	1.82	(0.54)	0.0337
drinking linear shape	0.42	(0.24)	0.0301
drinking quadratic shape	-0.56	(0.33)	0.0368
drinking average alter	1.24	(0.81)	0.0181

convergence *t* ratios all < 0.09 .

Overall maximum convergence ratio 0.19.

More structural effects

Default effects

Choose possible network effects for actor i , e.g.:
(others to whom actor i is tied are called here i 's 'friends')

- ① *out-degree effect*, controlling the density / average degree,

$$s_{i1}(x) = x_{i+} = \sum_j x_{ij}$$

- ② *reciprocity effect*, number of reciprocated ties

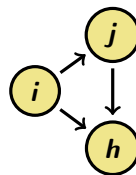
$$s_{i2}(x) = \sum_j x_{ij} x_{ji}$$

Four ways of closure (1)

Four potential effects representing network closure:

- ③ *transitive triplets effect*,
number of transitive patterns in i 's ties
($i \rightarrow j, j \rightarrow h, i \rightarrow h$)

$$s_{i3}(x) = \sum_{j,h} x_{ij} x_{jh} x_{ih}$$



transitive triplet

- ④ *transitive ties effect*,
number of actors j to whom i is tied indirectly
(through at least one intermediary: $x_{ih} = x_{hj} = 1$)
and also directly $x_{ij} = 1$),
 $s_{i4}(x) = \#\{j \mid x_{ij} = 1, \max_h (x_{ih} x_{hj}) > 0\}$

Four ways of closure (2)

- ⑤ *indirect ties effect*,
 number of actors j to whom i is tied indirectly
 (through at least one intermediary: $x_{ih} = x_{hj} = 1$)
 but not directly ($x_{ij} = 0$),
 = number of geodesic distances equal to 2,
 $s_{i5}(x) = \#\{j \mid x_{ij} = 0, \max_h (x_{ih} x_{hj}) > 0\}$



Four ways of closure (3)

- ⑥ *balance* or structural equivalence,
similarity between outgoing ties of i
with outgoing ties of his friends,

$$s_{i6}(x) = \sum_{j=1}^n x_{ij} \sum_{\substack{h=1 \\ h \neq i, j}}^g (1 - |x_{ih} - x_{jh}|) ,$$

[note that $(1 - |x_{ih} - x_{jh}|) = 1$ if $x_{ih} = x_{jh}$,
and 0 if $x_{ih} \neq x_{jh}$, so that

$$\sum_{\substack{h=1 \\ h \neq i, j}}^g (1 - |x_{ih} - x_{jh}|)$$

measures agreement between i and j .]



Four ways of closure (4)

Differences between these three network closure effects:

- transitive triplets effect: i more attracted to j
if there are *more* indirect ties $i \rightarrow h \rightarrow j$;
- transitive ties effect: i more attracted to j
if there is *at least one* such indirect connection ;
- balance effect:
 i prefers others j who make same choices as i .



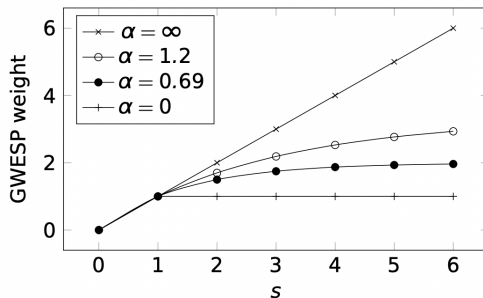
One way of closure: GWESP

Nowadays, we often use GWESP (geometrically weighted edgewise shared partners) - combines transTrip and transTies:

$$GWESP(i, \alpha) = \sum_j x_{ij} e^{\alpha} \left[1 - (1 - e^{-\alpha}) \overbrace{\sum_h x_{ih} x_{jh}}^{\# \text{com. partn.}} \right]$$

- for $\alpha \geq 0$ (effect parameter = $100 \times \alpha$).
- Default $\alpha = \log(2)$, parameter = 69

One way of closure: GWESP



Weight tie $i \rightarrow j$ for $s = \sum_h x_{ih}x_{jh}$

Degree-based effects

- 7 *in-degree related popularity effect*, sum friends' in-degrees

$$s_{i7}(x) = \sum_j x_{ij} \sqrt{x_{+j}} = \sum_j x_{ij} \sqrt{\sum_h x_{hj}}$$

related to dispersion of in-degrees

(can also be defined without the $\sqrt{}$ sign);

- 8 *out-degree related popularity effect*,

sum friends' out-degrees

$$s_{i8}(x) = \sum_j x_{ij} \sqrt{x_{j+}} = \sum_j x_{ij} \sqrt{\sum_h x_{jh}}$$

related to association in-degrees — out-degrees;

- 9 *Outdegree-related activity effect* ,

$$s_{i9}(x) = \sum_j x_{ij} \sqrt{x_{i+}} = x_{i+}^{1.5}$$

related to dispersion of out-degrees;

- 10 *Indegree-related activity effect* ,

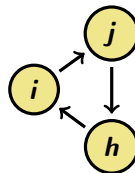
$$s_{i10}(x) = \sum_j x_{ij} \sqrt{x_{+i}} = x_{i+} \sqrt{x_{+i}}$$

related to association in-degrees — out-degrees;



Four ways of closure (5)

- 11 *three-cycle effect*,
 number of three-cycles in i 's ties
 $(i \rightarrow j, j \rightarrow h, h \rightarrow i)$
 $s_{i11}(x) = \sum_{j,h} x_{ij} x_{jh} x_{hi}$



three-cycle

This represents a kind of generalized reciprocity,
 and absence of hierarchy.

- 12 ... and potentially many others ...

More on selection effects

Selection effects

Preferences of actors dependent on their degrees:

- out ego - out alter degrees
- out ego - in alter degrees
- in ego - out alter degrees
- in ego - in alter degrees

All these are product interactions between the two degrees (or square roots).



Selection effects: types of evaluations

Four kinds of evaluation function effect associated with actor covariate v_i .

This applies also to behavior variables Z_h .

- 13 *covariate-related popularity*, 'alter'
sum of covariate over all of i 's friends
 $s_{i13}(x) = \sum_j x_{ij} v_j$
- 14 *covariate-related activity*, 'ego'
 i 's out-degree weighted by covariate
 $s_{i14}(x) = v_i x_{i+}$



Selection effects: similarity

- 15 *covariate-related similarity*,
sum of measure of covariate similarity
between i and his friends,

$$s_{i15}(x) = \sum_j x_{ij} \text{sim}(v_i, v_j)$$

where $\text{sim}(v_i, v_j)$ is the similarity between v_i and v_j ,

$$\text{sim}(v_i, v_j) = 1 - \frac{|v_i - v_j|}{R_V},$$

R_V being the range of V ;

- 16 *covariate-related interaction*, 'ego $\times$ alter'

$$s_{i16}(x) = v_i \sum_j x_{ij} v_j;$$



Selection effects: similarity

Snijders and Lomi (2019) *Beyond homophily: Incorporating actor variables in statistical network models*:

- for (non-binary) variables v_i
 - ▶ combination of tendencies of
 - homophily,
 - aspiration, and
 - social norm
 - ▶ yields 5 effects:
 - ① ego $x_{ij} v_i$
 - ② alter $x_{ij} v_j$
 - ③ ego-squared $x_{ij} v_i^2$
 - ④ ego-alter difference squared $x_{ij} (v_i - v_j)^2$ and
 - ⑤ alter squared $x_{ij} v_j^2$

Do we really *have* to use this?



Example van de Bunt (1)

Example (Gerhard van de Bunt)

Data

- 32 university freshmen (24 fem and 8 male)
- (here) 3 obs. (t_1 , t_2 , t_3) at 6, 9, and 12 weeks
- The relation: 'friendly relation'.

Missing entries $x_{ij}(t_m)$ set to 0 and not used in calculations of statistics.
Densities increase from 0.15 at t_1 via 0.18 to 0.22 at t_3 .

Example van de Bunt (2)

Example (Gerhard van de Bunt (cont.))

Very simple model: only out-degree and reciprocity effects

Effect	Model 1	
	par.	(s.e.)
Rate $t_1 - t_2$	3.51	(0.54)
Rate $t_2 - t_3$	3.09	(0.49)
Out-degree	-1.10	(0.15)
Reciprocity	1.79	(0.27)

rate parameters:

per actor about 3 opportunities for change between observations;

out-degree parameter negative:

on average, cost of friendship ties higher than their benefits;

reciprocity effect strong and highly significant ($t = 1.79/0.27 = 6.6$).

Example van de Bunt (3)

Example (Gerhard van de Bunt (cont.))

Evaluation function is

$$f_i(x) = \sum_j \left(-1.10 x_{ij} + 1.79 x_{ij} x_{ji} \right).$$

This expresses 'how much actor i likes the network'.

Adding a **reciprocated** tie (i.e., for which $x_{ji} = 1$) gives

$$-1.10 + 1.79 = 0.69.$$

Adding a **non-reciprocated** tie (i.e., for which $x_{ji} = 0$) gives

$$-1.10,$$

i.e., this has negative benefits.

Gumbel distributed disturbances are added:

these have variance $\pi^2/6 = 1.645$ and s.d. 1.28.

Example van de Bunt (4)

Example (Gerhard van de Bunt: with simple closure)

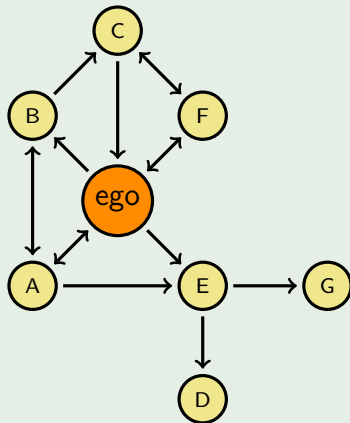
The estimates with only transitive ties:

Structural model with one network closure effect

Effect	Model 3	
	par.	(s.e.)
Rate $t_1 - t_2$	3.89	(0.60)
Rate $t_2 - t_3$	3.06	(0.47)
Out-degree	-2.14	(0.38)
Reciprocity	1.55	(0.28)
Transitive ties	1.30	(0.41)

Example van de Bunt (5)

Example (Gerhard van de Bunt: with simple closure (cont.))



for ego:

out-degree $x_{i+} = 4$

$\#\{\text{recipr. ties}\} = 2,$

$\#\{\text{trans. ties}\} = 3.$

Example van de Bunt (6)

Example (Gerhard van de Bunt: with simple closure (cont.))

The evaluation function is

$$f_i(x) = \sum_j \left(-2.14 x_{ij} + 1.55 x_{ij} x_{ji} + 1.30 x_{ij} \max_h (x_{ih} x_{hj}) \right)$$

(note: $\sum_j x_{ij} \max_h (x_{ih} x_{hj})$ is $\#\{\text{trans. ties}\}$)
so its current value for this actor is

$$f_i(x) = -2.14 \times 4 + 1.55 \times 2 + 1.30 \times 3 = -1.56.$$



Example van de Bunt (7)

Example (Gerhard van de Bunt: with simple closure (cont.))

Options when 'ego' has opportunity for change:

	out-degr.	recipr.	trans. ties	gain	prob.
current	4	2	3	0.00	0.061
new tie to C	5	3	5	+2.01	0.455
new tie to D	5	2	4	+0.46	0.096
new tie to G	5	2	4	+0.46	0.096
drop tie to A	3	1	0	-3.31	0.002
drop tie to B	3	2	1	-0.46	0.038
drop tie to E	3	2	2	+0.84	0.141
drop tie to F	3	1	3	+0.59	0.110

The actor adds random influences to the gain (with s.d. 1.28), and chooses the change with the highest total 'value'.

Example van de Bunt (8)

Example (Gerhard van de Bunt: with more closure)

Effect	Model 3	
	par.	(s.e.)
Rate $t_1 - t_2$	4.64	(0.80)
Rate $t_2 - t_3$	3.53	(0.57)
Out-degree	-0.90	(0.58)
Reciprocity	2.27	(0.41)
Transitive triplets	0.35	(0.06)
Transitive ties	0.75	(0.45)
Three-cycles	-0.72	(0.21)
In-degree popularity ($\sqrt{}$)	-0.71	(0.27)

Conclusions:

Reciprocity, transitivity;
negative 3-cycle effect;
negative
popularity effect.

Example van de Bunt (9)

Example (Gerhard van de Bunt: Add effects of gender & program, smoking similarity)

Effect	Model 4	
	par.	(s.e.)
Rate $t_1 - t_2$	4.71	(0.80)
Rate $t_2 - t_3$	3.54	(0.59)
Out-degree	-0.81	(0.61)
Reciprocity	2.14	(0.45)
Transitive triplets	0.33	(0.06)
Transitive ties	0.67	(0.46)
Three-cycles	-0.64	(0.22)
In-degree popularity (✓)	-0.72	(0.28)
Sex (M) alter	0.52	(0.27)
Sex (M) ego	-0.15	(0.27)
Sex similarity	0.21	(0.22)
Program similarity	0.65	(0.26)
Smoking similarity	0.25	(0.18)

Conclusions:

Trans. ties now
not needed any more
to represent
transitivity;
men more popular;
program similarity.

Example van de Bunt (10)

Example (Gerhard van de Bunt: selection table)

We may do the calculations with $F = 0$, $M = 1$ (even if centered) The joint effect of the gender-related effects for the tie variable x_{ij} from i to j is

$$-0.15 z_i + 0.52 z_j + 0.21 I\{z_i = z_j\}.$$

$i \setminus j$	F	M
F	0.21	0.52
M	-0.15	0.58

Conclusion:
men seem not to like female friends...?

More on influence effects



Influence effects

Many different reasons why networks are important for behavior:

- ① *imitation* :
individuals imitate others
(basic drive; uncertainty reduction).
- ② *social capital* :
individuals may use resources of others;
- ③ *coordination* :
individuals can achieve some goals
only by concerted behavior;

In this presentation, only imitation is considered, but the other two reasons are also of eminent importance.



Influence effects

Basic effects for dynamics of behavior f_i^Z :

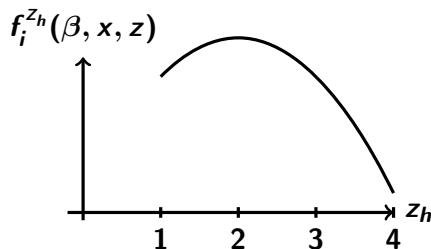
$$f_i^Z(\beta, x, z) = \sum_{k=1}^L \beta_k s_{ik}(x, z),$$

- ① *tendency* ,
 $s_{i1}^Z(x, z) = z_{ih}$
- ② *quadratic tendency*, 'effect behavior on itself',
 $s_{i2}^Z(x, z) = z_{ih}^2$
Quadratic tendency effect important for model fit.



Influence effects

For a negative quadratic tendency parameter, the model for behavior is a unimodal preference model.



For positive quadratic tendency parameters, the behavior objective function can be bimodal ('positive feedback').

Influence effects

- ③ *behavior-related average similarity*,
 average of behavior similarities between i and friends
 $s_{i3}(x) = \frac{1}{x_{i+}} \sum_j x_{ij} \text{sim}(z_{ih}, z_{jh})$
 where $\text{sim}(z_{ih}, z_{jh})$ is the similarity between v_i and v_j ,

$$\text{sim}(z_{ih}, z_{jh}) = 1 - \frac{|z_{ih} - z_{jh}|}{R_{Z^h}},$$

R_{Z^h} being the range of Z^h ;

- ④ *average behavior alter* — an alternative to similarity:
 $s_{i4}(x, z) = z_{ih} \frac{1}{x_{i+}} \sum_j x_{ij} z_{jh}$

Effects 3 and 4 are alternatives for each other:
 they express the same theoretical idea of influence
 in mathematically different ways.

The data will have to differentiate between them.



Influence effects

Network position can also have influence on behaviour dynamics
e.g. through degrees rather than through behaviour
of those to whom one is tied:

- 5 *popularity-related tendency*, (in-degree)

$$s_{i5}(x, z) = z_{ih} x_{+i}$$

- 6 *activity-related tendency*, (out-degree)

$$s_{i6}(x, z) = z_{ih} x_{i+}$$



Influence effects

- 7 *dependence on other behaviours* ($h \neq \ell$) ,
 $s_{i7}(x, z) = z_{ih} z_{i\ell}$

For both the network and the behaviour dynamics,
extensions are possible depending on the network position.



Influence effects

The *similarity effect* in evaluation function :

sum of absolute behaviour differences between i and his friends

$$s_{i2}(x, z) = \sum_j x_{ij} \text{sim}(z_{ih}, z_{jh}) .$$

This is fundamental both

to network selection based on behaviour,

and to behavior change based on network position.



Influence effects: Example

Example (Smoke rings)

Study of smoking initiation and friendship

(following up on earlier work by P. West, M. Pearson & others).

One school year group from a Scottish secondary school starting at age 12-13 years, was monitored over 3 years; total of 160 pupils, of which 129 pupils present at all 3 observations; with sociometric & behavior questionnaires at three moments, at appr. 1 year intervals.

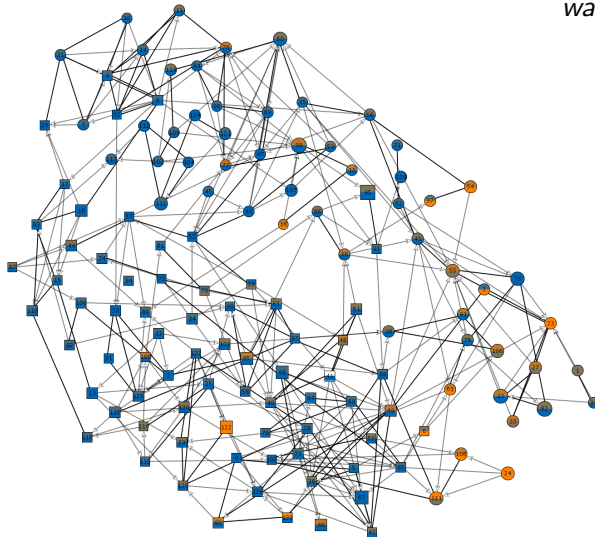
Smoking: values 1–3;

drinking: values 1–5;

covariates:

gender, smoking of parents and siblings (binary),

money available (range 0–40 pounds/week).



wave 1

girls: circles

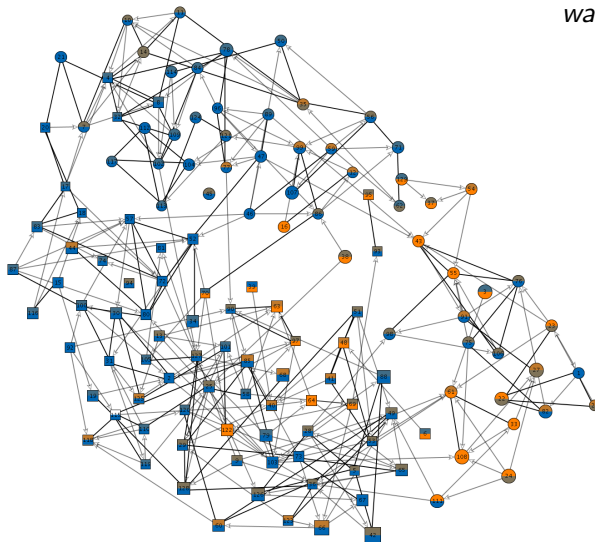
boys: squares

node size: pocket money

color: top = drinking

bottom = smoking

(orange = high)



wave 2

girls: circles

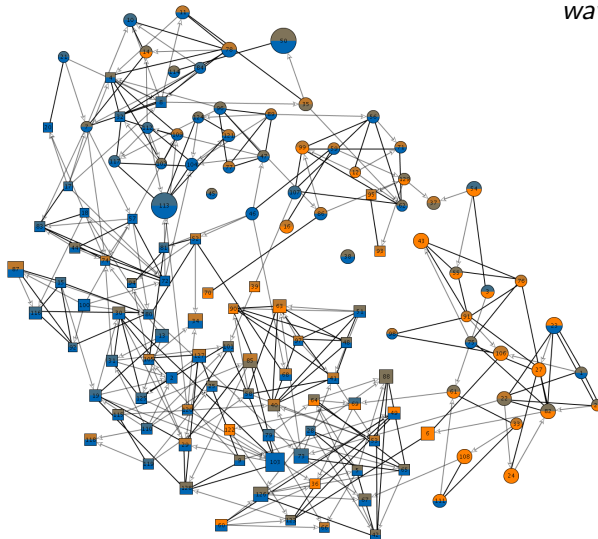
boys: squares

node size: pocket money

color: top = drinking

bottom = smoking

(orange = high)



wave 3

girls: circles

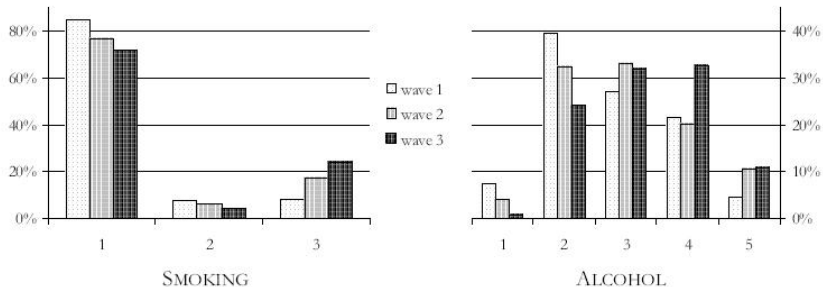
boys: squares

node size: pocket money

color: top = drinking

bottom = smoking

(orange = high)

FIGURE 2. — OBSERVED DISTRIBUTION OF SUBSTANCE USE IN THE THREE WAVES.

More realistic model

<i>Friendship dynamics</i>	Rate 1	18.67	(2.17)
	Rate 2	12.42	(1.30)
	Outdegree	-1.57	(0.27)
	Reciprocity	2.04	(0.13)
	Transitive triplets	0.35	(0.04)
	Transitive ties	0.84	(0.09)
	Three-cycles	-0.41	(0.10)
	In-degree based popularity ($\checkmark$)	0.05	(0.07)
	Out-degree based popularity ($\checkmark$)	-0.45	(0.16)
	Out-degree based activity ($\checkmark$)	-0.39	(0.07)
	Sex alter	-0.14	(0.08)
	Sex ego	0.08	(0.10)
	Sex similarity	0.66	(0.08)
	Romantic exp. similarity	0.10	(0.06)
	Money alter (unit: 10 pounds/w)	0.11	(0.05)
	Money ego	-0.06	(0.06)
	Money similarity	0.98	(0.27)

More realistic model (continued)

<i>Friendship dynamics</i>	Drinking alter	−0.01	(0.07)
	Drinking ego	0.09	(0.09)
	Drinking ego × drinking alter	0.14	(0.06)
	Smoking alter	−0.08	(0.08)
	Smoking ego	−0.14	(0.09)
	Smoking ego × smoking alter	0.03	(0.08)

<i>Smoking dynamics</i>	Rate 1	4.74	(1.88)
	Rate 2	3.41	(1.29)
	Linear tendency	-3.39	(0.45)
	Quadratic tendency	2.71	(0.40)
	Ave. alter	2.00	(0.95)
	Drinking	-0.11	(0.24)
	Sex (F)	-0.12	(0.35)
	Money	0.10	(0.20)
	Smoking at home	-0.05	(0.29)
	Romantic experience	0.09	(0.33)

<i>Alcohol consumption dynamics</i>	Rate 1	1.60	(0.32)
	Rate 2	2.50	(0.42)
	Linear tendency	0.44	(0.17)
	Quadratic tendency	-0.64	(0.22)
	Ave. alter	1.34	(0.61)
	Smoking	0.01	(0.21)
	Sex (F)	0.04	(0.22)
	Money	0.17	(0.16)
	Romantic experience	-0.19	(0.27)

Conclusion:

In this case, the conclusions from a more elaborate model – i.e., with better control for alternative explanations – are similar to the conclusions from the simple model.

There is evidence for friendship selection based on drinking, and for social influence with respect to smoking and drinking.



Parameter interpretation for behaviour change

Omitting the non-significant parameters yields the following objective functions.

For smoking

$$f_i^{z_1}(\hat{\beta}, x, z) = -3.39(z_{i1} - \bar{z}_1) + 2.71(z_{i1} - \bar{z}_1)^2 + 2.00(z_{i1} - \bar{z}_1)(\bar{z}_{i1,fr} - \bar{z}_1),$$

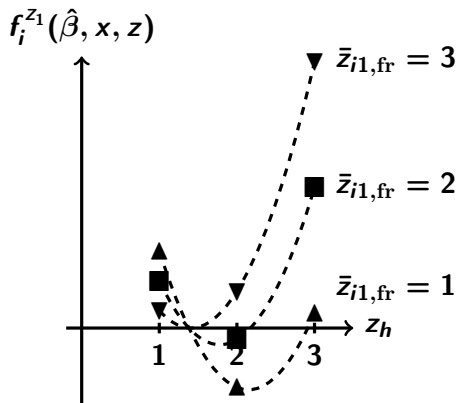
where z_{i1} is smoking of actor i : values 1–3, mean 1.4.

$\bar{z}_{i1,fr}$ is the average smoking behavior of i 's friends.

Convex function – consonant with addictive behavior.



$$-3.39(z_{i1} - \bar{z}_1) + 2.71(z_{i1} - \bar{z}_1)^2 + 2.00(z_{i1} - \bar{z}_1)(\bar{z}_{i1,fr} - \bar{z}_1)$$



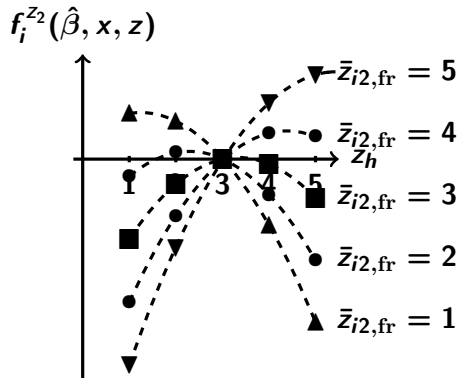
For drinking the objective function (significant terms only) is

$$f_i^{z_2}(\hat{\beta}, x, z) = \\ 0.44 (z_{i2} - \bar{z}_2) - 0.64 (z_{i2} - \bar{z}_2)^2 + 1.34 (z_{i2} - \bar{z}_2) (\bar{z}_{i2, \text{fr}} - \bar{z}_2) ,$$

where z_{i2} is drinking of actor i : values 1–5, mean 3.0.

Unimodal function – consonant with non-addictive behavior.

$$0.44(z_{i2} - \bar{z}_2) - 0.64(z_{i2} - \bar{z}_2)^2 + 1.34(z_{i2} - \bar{z}_2)(\bar{z}_{i2,fr} - \bar{z}_2)$$



Testing parameters using score-type test

We might not be able to fit everything (no, really ...)

How test multiple parameters without estimation?

Score (-type) test

If MoM estimate, then

$$\hat{z} - z \approx 0$$

If this holds for the statistic z_K for a parameter $\theta_K = 0$, then $\theta_K = 0$ is a good value

Test non-estimated parameters θ^* , with the statistic

$$(\hat{z} - z)^\top D(\theta^*)^{-1}(\hat{z} - z)$$

where $\hat{z}$ is a vector of expected statistics for parameters θ^* , and D is a suitably scaled covariance matrix.



Trouble shooting: non-convergence

What stochastic approximation algorithm does

- ① Gauging sensitivity of (estimation) statistics Z to parameters θ ;
- ② Robbins-Monro updates for θ
 - ▶ *nsub* subphases (usually 4)
 - ▶ decreasing step sizes, determined by *firstg*
- ③ Final: *n3* runs, θ constant at $\hat{\theta}$
 - ▶ Check deviations from targets

$$E_{\hat{\theta}}\{Z\} - z$$

- ▶ estimating standard errors



Trouble shooting: non-convergence - bad start!

Initial values:

① `sienaAlgorithmCreate`

- ▶ `useStdInits=FALSE`: parameter values in effects object
 - ◉ starting with standard values
 - ◉ can be modified by functions `setEffect` and `updateTheta`
- ▶ `useStdInits=TRUE`:
 - ◉ standard initial values
 - ◉ the values put in the effects object by `getEffects`.

② With arg `prevAns` passed to `siena07`

- ▶ initial values used from existing `sienaFit` object,
- ▶ Skipping Phase 1 if mods identical



Trouble shooting: non-convergence - when?

Standard initial values mostly fine but for

- non-directed networks
- two-mode networks
- monotonic dependent variables
- multivariate networks with constraints
- data sets with many structurally determined values.

You may try

- start with only rate and density (-effects)
- `updateTheta` $\Rightarrow$ restart



Trouble shooting: non-convergence - normal

Typically, for `tconv.max` > 0.25 ,

- repeat estimation,
- using the `prevAns` parameter in `siena07`,
- until `tconv.max` < 0.25

Warning sign

- estimation *diverges right away*
 - ▶ check data and model specification;
 - ▶ perhaps use a simpler model.
- estimation still *diverges right away*, either:
 - ▶ estimate a simpler model, and use the result for `prevAns` with the intended model, or
 - ▶ use a smaller value for `firstg` (default: 0.2)

NB: `siena07` will **tell you** if effects *co-linear* - so don't worry about that



Trouble shooting: non-convergence - brute force

If model resits converging ($tconv.max > 0.25$ after many restarts)

- Brute force: increase e.g. $n2start$ and/or $n3$, with smaller $firststg$
- Better model
- Check for time-heterogeneity
- Better data
 - ▶ Do you miss important covariates?
 - ▶ Do your variables need transformations?
 - ▶ etc

